

CSCE 671 HUMAN-COMPUTER INTERACTION · MODULE 13 · LECTURE NOTES

Claiming Usability Honestly

What a study establishes.

MODULE OUTCOMES

State what each method establishes.

Identify the standard overclaims.

Write a defensible usability claim.

Place this course relative to CSCE 679 and CSCE 632.

State a proportionate practice for real design work.

1 What each method establishes

Method	What it establishes	What it does not
Heuristic review	That these principles are violated in these places.	That users will actually struggle (Module 05 §4).
Usability test, n = 5	That these problems exist, and the mechanism behind each.	Any rate, and any comparison between designs (Module 07 §2).
Controlled experiment	A measured difference, under these conditions, with this interval.	Long-term behaviour, or whether the novelty wears off (Module 06 §4, Module 08 §4).
Interview	What they report, and how they explain their own behaviour.	What they actually do (Module 09 §3).
Observation	What they do, and the context it happens in.	Frequencies in a population.
Analytics	What happened, at scale, across real users.	Why it happened — which needs Module 09.

The last two rows are complementary rather than redundant — analytics tells you *what* and qualitative work tells you *why*, and neither substitutes for the other: a drop-off at step three is a finding with no explanation, and five observed sessions explain a mechanism with no magnitude. **The does-not-establish column is this module's content**, and it is the one usually omitted from a methods summary.

“WE TESTED IT” IS NOT A CLAIM UNTIL YOU SAY WHICH METHOD AND WHAT IT SHOWED

Each row above supports a narrow and specific claim — and **'we tested it and users liked it'** names no method, no number of participants, no tasks, and no measure, which makes it unassessable.

And **'users liked it'** is the weakest possible finding in any case, because a **participant in a room with the designer will say they liked almost anything** (Module 06 §4's expressed-interest problem, and Module 08 §4's demand characteristics).

So the honest form names the method, the number of participants, the tasks, and what actually happened — **four clauses, all of them cheap to supply** and all of them making the claim checkable.

Which is the same correction that CSCE 701 Module 13, CSCE 713 Module 13, and CSCE 670 Module 13 all make — **substituting an activity for a finding** — and **this program has now identified it in four different subjects**, which suggests it is the general failure rather than a disciplinary quirk.

2 Overclaims, and defensible claims

The claim	The correction
'Users liked it.'	How many users, asked how, and compared against what? And liking is not usability.
'It is intuitive.'	Intuitive to whom? Intuition is prior exposure (Module 04 §2) — see the note.
'80% of users prefer it.'	From five participants? That is four people, and the percentage implies a precision the sample cannot carry.
'The redesign improved usability.'	On which tasks, measured how, and what got worse? (Project 2's regression requirement.)
'It is accessible.'	Tested with whom, against which standard, with what assistive technology? (CSCE 632.)
'Best practice says...'	Which study, in which context, with what effect size? A great deal of 'best practice' is a single old study.

'It is intuitive' is the one to attack — **intuition is prior exposure** (Module 04 §2's conventions), **so the claim is always relative to a population and is usually relative to the designer's own**. Nothing is intuitive in the abstract: a scroll bar, a hamburger menu, and a pinch gesture all had to be learned. **The intuition correction is the most generally useful one in the table**, because the word is used constantly and is never qualified.

- **'Five participants attempted three tasks; four of them failed task two at the same step, which we attributed to a missing signifier on the filter control.'** Method, number, task, step, and mechanism — all five checkable.
- **'After the change, five new participants completed task two; one took notably longer and we do not know why.'** Including the unexplained case, which is what distinguishes a report from a success story.

- **'In a counterbalanced within-subject study with twelve participants, design B was 18% faster on this task (95% confidence interval 4 to 32%).'** Design, n, effect size, and interval (Module 08 §3).
- **'Our participants were twelve colleagues; our users are not twelve colleagues.'** The sampling limitation, stated rather than implied (Module 08 §4).
- **And: 'we have not tested with screen reader users, on mobile, or across more than one session.'** The last one is the limitation statement, and it is Project 2's hardest-graded requirement — because it requires knowing what your method structurally cannot reach, which is §1's right-hand column applied to your own work.

3 The semester

Course	What it covers	Its central correction
CSCE 671 (this one)	Interaction — how people use what you built.	Your judgement of your own design is unreliable, so watch somebody (Module 01 §2).
CSCE 679	Visual representation of data.	The encoding determines what can be read from a picture — so the choice is not aesthetic.
CSCE 632	Access — who can use it at all.	Designing for a constraint improves the design for everybody, which is a quality argument rather than a fairness one.

All three courses correct the same error: designing for an imagined user who resembles the designer — in their knowledge (671), in their perception (679), and in their capabilities (632). And all three answer it the same way, with observation rather than with argument: watch somebody, test the encoding, try it with the assistive technology. The shared correction is the semester's conclusion, and it is more durable than any individual principle in it.

4 A proportionate practice

- 1 · Watch five people (Module 07) — which costs a day, needs no equipment, and is the only reliable source of information in this subject (Module 01 §2) — and almost nobody does it.
- 2 · Fix the failures mechanically, attributing each to one of Module 01 §1's six mechanisms before proposing anything, so that each proposal is testable.
- 3 · Check the defaults, the error messages, and the feedback placement (Modules 05, 10, and 02) — all three cheap, all three high-return, and all three fixable without a redesign.
- 4 · Run the accessibility review (CSCE 632) — which finds problems the five participants structurally could not have found, and which is the single largest gap in a five-participant study.
- 5 · And only then measure a difference (Module 08), which is expensive and answers a much narrower question than the earlier steps do. Step 1 before everything is the ordering that matters — and it is the step that gets deferred because it is socially uncomfortable rather than because it is expensive, which is worth naming so that the discomfort can be acknowledged rather than rationalised.

WHERE THIS COURSE LEAVES YOU

You can diagnose an interface failure mechanically — a missing signifier, absent feedback, a perceptual failure, an attention failure, a memory demand, or a model mismatch — **rather than describing it as 'confusing'**, which is the difference between a finding and an impression.

You can watch five people without helping, and analyse what you saw into mechanisms and severities — **which is the skill that makes the rest of the course evidence rather than argument** (Module 01 §2's governing constraint).

And you know that all of this applies to the interfaces you build for other programmers (Module 10) — **APIs, error messages, configuration, and tooling** — **which is most of what you will actually design** in a career in this field.

The closing rule is the program's, unchanged across thirty-one courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it means naming the method, the participants, and the tasks — because **'it is intuitive'** names your own prior exposure and nothing else at all.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lazar, Feng & Hochheiser, the reporting chapters	§1 and §2 — what each method supports, and how to report it. Library copy.
Cockton — on the limits of usability claims (free via the CHI proceedings)	§2's overclaims, in the field's own literature — and the limitations sections of CHI papers are the model to follow.
Greenberg & Buxton — Usability Evaluation Considered Harmful (free)	§1's first two rows argued sharply — when evaluation is the wrong activity, which is a useful corrective.
Dragicevic — Fair Statistical Communication in HCI (free)	§2's third claim form — intervals rather than significance, and why.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 For each of the six methods**, write what it establishes and what it does not.
- 2 Find a usability claim** made about a product and assess it against §2's table.
- 3 Find 'intuitive' used without qualification** and supply the missing population.

- 4 **Rewrite one of your own findings** in §2's claim form.
- 5 **Write your limitation statement** for Project 2.
- 6 **Name the correction** each of the three Semester 11 courses makes.
- 7 **Audit your own practice** against §4's five steps.
- 8 **Determine why step 1 has not happened**, honestly.
- 9 **Revisit Module 01's last exercise** and compare.
- 10 **Project 2 is now due.** Submit the redesign, the before and after sessions with different participants, the per-participant results, the qualitative account, the accessibility review, and the limitation statement.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 For six methods, state what each establishes and does not.
- 2 Why are analytics and qualitative work complementary?
- 3 Why is 'we tested it' not a claim?
- 4 Why is 'users liked it' the weakest finding?
- 5 Give six overclaims and the correction to each.
- 6 Why is 'intuitive' the one to attack?
- 7 Give four defensible claim forms.
- 8 What error do all three Semester 11 courses correct, and how?
- 9 Give the five-step practice in order.
- 10 Why does step 1 get deferred?